DELHI PUBLIC SCHOOL JANKIPURAM, LUCKNOW

SESSION 2025-2026: HALF YEARLY EXAMINATION

SET–I

Subject: Data Science (419)

TIME : 2 Hrs. M.M. : 50

Date 26/09/2025 Name : Siddharth Parth Sarthi Class/Sec X "D"

General Instructions:

Read the following instructions carefully and follow them:

(i) This question paper consists of 21 questions in two sections : Section A and Section B.

(ii) Section A has Objective Type Questions, whereas Section B contains Subjective Type Questions.

(iii) Out of the given (5 + 16 =) 21 questions, the candidate has to answer (5 + 10 =) 15 questions in the allotted (maximum) time of 2 hours.

(iv) All questions of a particular section must be attempted in the correct order.

(v) 1. Section A : Objective Type Questions (24 marks) :

(a) This section has 5 questions.

(b) There is no negative marking.

(c) Do as per the instructions given.

(d) Marks allotted are mentioned against each question/part.

(vi) Section B : Subjective Type Questions (26 marks) :

(a) This section has 16 questions.

(b) A candidate has to do 10 questions.

(c) Do as per the instructions given.

(d) Marks allotted are mentioned against each question/part.

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Section A

(Objective Type Questions) (24 marks)

1. **Answer any 4 out of the given 6 questions on Employability Skills.** 4 X 1 = 4

(i) In written communication, which element is MOST important?
a) Handwriting clarity
b) Punctuation
c) Grammar
d) All of these

(ii) Which of the following is an example of cloud storage?
a) Google Drive
b) USB Pen Drive
c) External Hard Disk
d) CD/DVD

(iii) Which file extension is used for MS Word documents?
a) .xls
b) .ppt

c) .docx
d) .pdf

(iv) Which of these is an example of **positive motivation?**
a) Fear of punishment
b) Hope of reward
c) Peer pressure
d) Stress

(v) The process of sending and receiving messages between two or more people is called:
a) Transmission
b) Communication
c) Interpretation
d) Networking

(vi) Stress management techniques include:
a) Meditation
b) Exercise
c) Proper rest
d) All of the above

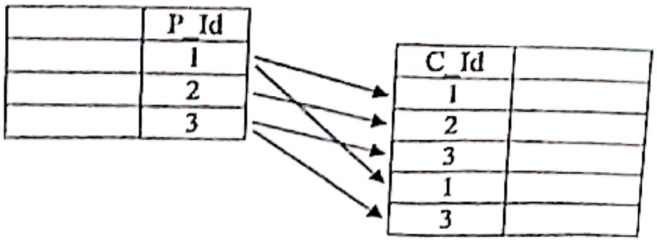
2. **Answer any 5 out of the given 6 questions.** **5 X 1=5**

(i) Identify the correct pair:
a) Mean → Data spread
b) Median → Middle value
c) Mode → Average of data
d) Variance → Middle value

(ii) The type of bias where certain groups are left out of the survey is called:
a) Response bias
b) Selection bias
c) Measurement bias
d) Interviewer bias

(iii) Z-score measures:
a) Central tendency
b) Standard deviation
c) Number of observations
d) Distance from the mean in terms of standard deviation

(iv) Identify the type of join from the following diagram.



(a) One to One
(b) One to Many
(c) Many to Many

(d)Many to One

(v) Identify the type of data: "Marks obtained by students in Mathematics."
a) Qualitative
b) Discrete numerical
c) Continuous numerical
d) Categorical

(vi) A dataset has values: 2, 4, 6, 8, 10. Its mean is:
a) 4
b) 5
c) 6
d) 8

3. **Answer any 5 out of the given 6 questions.** **5 X 1= 5**

(i) A bar graph is best used for representing:
a) Continuous data
b) Categorical data
c) Both a and b
d) None of these

(ii) A dataset has values: 2, 4, 6, 9, 14. Its median is:
a) 4
b) 5
c) 6
d) 7

(iii) Quartiles of a dataset partition the data in _____ equal parts.
a) One
b) Two
c) Three
d) Four

(iv) The middle value of an ordered dataset is called:
a) Mean
b) Mode
c) Median
d) Range

(v) Which data type is represented by "Temperature in Celsius"?
a) Numerical
b) Categorical
c) Ordinal
d) None of the above

(vi) A pie chart is generally used for:
a) Showing proportions of categories
b) Representing time series data
c) Displaying frequency distribution
d) Plotting continuous variables

4. **Answer any 5 out of the given 6 questions.** **5 X 1= 5**

(i) Problem-solving is best described as:
a) Avoiding challenges
b) Finding solutions to difficult situations
c) Postponing work
d) Ignoring issues

(ii) Which of the following is an important quality of teamwork?
a) Cooperation
b) Isolation
c) Competition
d) Domination

(iii) Adaptability refers to:
a) Taking responsibility
b) Adjusting to new situations smoothly
c) Following old habits
d) Ignoring new challenges

(iv) Z-score calculation needs:
a) Only count
b) Mean and standard deviation
c) Only the median
d) Mode only

(v) Probability helps in identifying:
a) Certainty only
b) Likelihood of events
c) Exact results
d) Random decisions

(vi) Distribution helps in:
a) Identifying central values only
b) Understanding variability and patterns
c) Removing outliers only
d) Simplification only

5. **Answer any 5 out of the given 6 questions.** **5 X 1= 5**

(i) The median is:
a) The middle value
b) Average of values
c) Most frequent value
d) Sum of values

(ii) A two-way frequency table helps in representing:
a) Single variable
b) Relationship between two categories
c) Trend over time
d) Random data

(iii) The median of the data set (5, 9, 15, 23, 34, 43) is
a) 15
b) 19
c) 20
d) 23

(iv) A subset of data refers to:
a) The entire data set
b) A portion of the dataset
c) Duplicate records
d) Aggregated summary

(v) A statistical problem-solving process includes:
a) Arbitrarily selecting data
b) Defining problem, collecting, analyzing, and interpreting data
c) Ignoring patterns
d) Not validating results

(vi) The probability of getting a red coloured queen in a deck of playing cards is:
a) 1/13
b) 1/26
c) 1/52
d) None of the above

SECTION B

(Subjective Type Questions) *(26 marks)*

Answer any 3 out of the given 5 questions on Employability Skills. Answer each question in 20 - 30 words. 3 X 2 = 6

6. Why is feedback important in communication? complete
7. Briefly explain any two stress management techniques.
8. Mention any two advantages of using cloud storage.
9. State two ways to build self-confidence.
10. List two precautions you should take while browsing the internet.

Answer any 4 out of the given 6 questions in 20 - 30 words each. 4 X 2 = 8

11. Explain the term **outlier** with an example.
12. A survey was conducted in a class of 10 students regarding the number of books read last month:
    Data = {2, 3, 5, 4, 3, 6, 7, 2, 5, 1}.
    i) Find the mean number of books read.
    ii) Interpret the result.
13. A dataset has mean = 50, standard deviation = 5. Calculate the z-score of a value = 60.
14. Briefly explain any two important applications of Data Science in real life.
15. Define (i) Quartile (ii) Decile
16. Explain "survivor bias" with a real-life example.

Answer any 3 out of the given 5 questions in 50 - 80 words each. 3 X 4 = 12

17. A company surveyed the monthly mobile data usage (in GB) of 10 employees:
    Data = {12, 15, 20, 18, 10, 22, 25, 15, 18, 20}.

    i) Construct the frequency table (classes: 10–14, 15–19, 20–24, 25–29).
    ii) Represent the data using a histogram.
    iii) Comment on the distribution pattern (Skewness).

18. The marks scored by 12 students are:
5, 7, 10, 15, 20, 25, 28, 30, 35, 40, 42, 50.
Find the **7th decile (D_7)**.

19. Given Data Set {12, 17, 25, 34, 46, 55, 68, 73}, calculate the following
    (i) Median
    (ii) IQR

20. Define
    (i) Selection Bias
    (ii) Recall Bias

21. Differentiate between
    (i) Continuous and discrete data
    (ii) Variance and Standard Deviation

$\frac{i \times (n+1)}{10}$

$i = 7, n =$

$d_i =$

$\frac{7 \times (12+1)}{10}$

35 and 40

$35 + 0.1(40-35)$

$35 + 0.1(5)$

$35 + 0.5 = 35.5$